IUT de Paris - Rives de Seine
Université Paris Cité

R4.Real.11 – Développement pour applications mobiles

– TP 4 –

Menus déroulant en se basant sur des fragments

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Résumé

Prérequis

- Gestion des vues et des layouts
- Gestion des évènements

Objectifs du TP

- Afficher une liste d'éléments
- Manipuler des fragments
- Créer un menu déroulant (Drawer)

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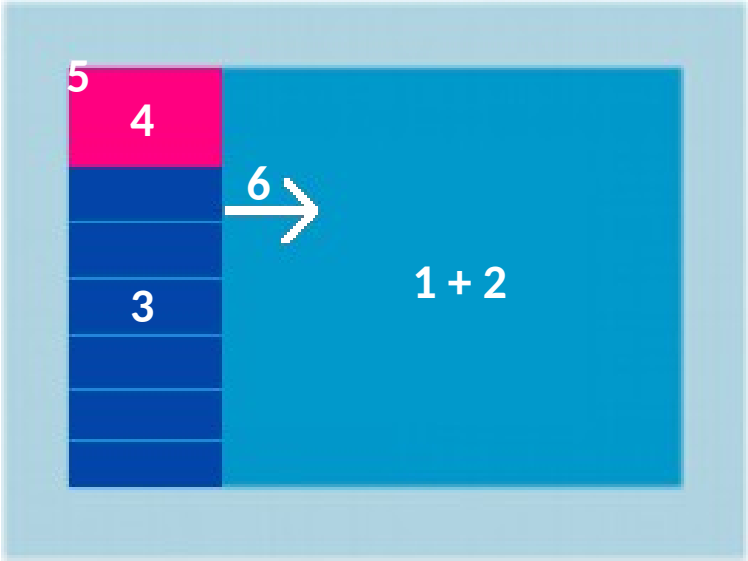
DrawerLayout / NavigationView



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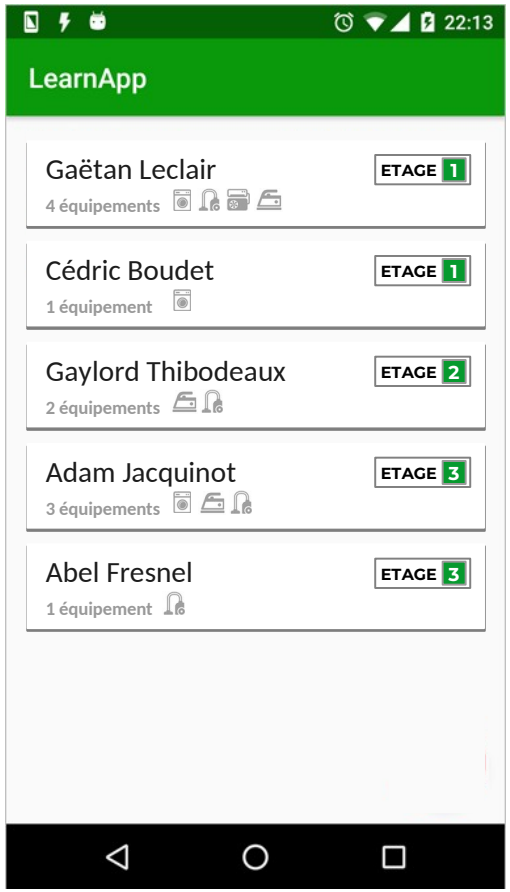
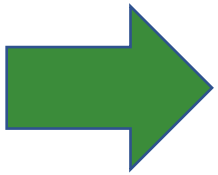
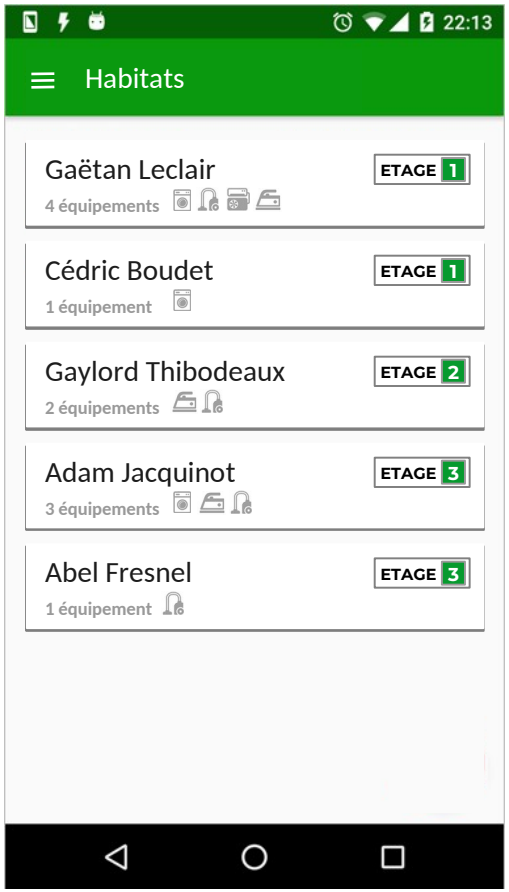
Activité simple vs. Activité avec menu déroulant







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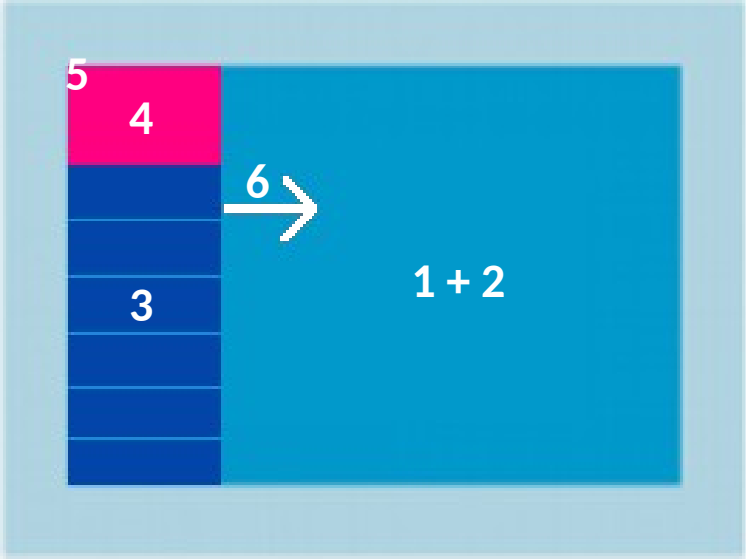
DrawerLayout / NavigationView

Etapes de création

1. Créer un fragment pour chaque élément du menu
2. Créer un layout pour chaque fragment
   - Importer la bibliothèque **Android Design Support**
3. Créer un menu d'éléments pour le **`NavigationView`**
4. Créer un layout pour l'entête du **`NavigationView`**
5. Ajouter DrawerLayout dans la racine du **layout de l'activité**
   - `androidx.drawerlayout.widget.DrawerLayout`
   - `com.google.android.material.navigation.NavigationView`
6. Instancier le **`DrawerLayout`** et intercepter les évènements **de clic**



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DrawerLayout / NavigationView

1. Créer un fragment pour chaque page

/java/FirstFragment.java

```
public class FirstFragment extends Fragment{
   ...
   public FirstFragment(){ ... }

   @Override
   public View onCreateView(LayoutInflater inflater,
                            ViewGroup container,
                            Bundle savedInstanceState) {
      getActivity().setTitle("First title");
      View rootView=inflater.inflate(R.layout.fragment_first, container, false);
      ...

      return rootView;
   }
}
```


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DrawerLayout / NavigationView

2a. Créer un layout pour chaque fragment

/res/layout/fragment_first.xml

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <!-- content -->

</LinearLayout>
```


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DrawerLayout / NavigationView

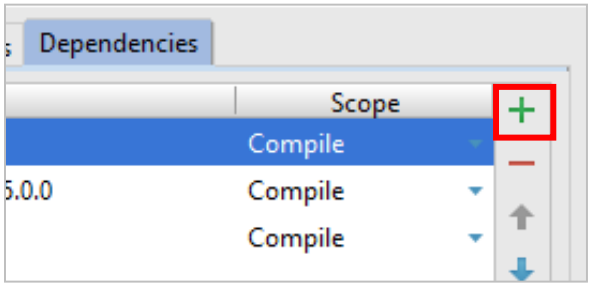
2b. Importer Android Design Support

Bibliothèque :

- `androidx.drawerlayout:drawerlayout:1.1.1`
- `com.google.android.material:material:1.4.1`

Méthode 1 :

- Project Structure > Modules > app
  > Onglet Depenendencies



Méthode 2 :

/Gradle Scripts/build.gradle (Module: app)

```
dependencies {
    ...
    implementation ("androidx.drawerlayout:drawerlayout:1.1.1")
    Implementation ("com.google.android.material:material:1.4.1")
}
```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DrawerLayout / NavigationView

3. Créer un menu d'éléments

/res/menu/nav_menu.xml

```
<menu xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto">

   <item
       android:id="@+id/nav_first"
       android:icon="@drawable/ic_menu_first"
       android:title="@string/first_item" />
   <item
       android:id="@+id/nav_second"
       android:icon="@drawable/ic_menu_second"
       android:title="@string/second_item" />
   ...
</menu>
```


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4. Créer un layout pour l'entête

/res/layout/nav_header.xml

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="horizontal">

    <!-- content -->

</LinearLayout>
```


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DrawerLayout / NavigationView

5. Ajouter `DrawerLayout` dans la racine

/res/layout/activity_main.xml

```
<androidx.drawerlayout.widget.DrawerLayout
    android:id="@+id/drawer" ... >

    <FrameLayout
            android:id="@+id/contentFL"
            android:layout_width="match..."
            android:layout_height="match..." />

    <com.google.android.material.navigation.NavigationView
        android:id="@+id/nav_view"
        android:layout_width="wrap..."
        android:layout_height="match..."
        android:layout_gravity="start"
        app:headerLayout="@layout/nav_header"
        app:menu="@menu/nav_menu" />

</<androidx.drawerlayout.widget.DrawerLayout>
```


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DrawerLayout / NavigationView

6. Instancier le `DrawerLayout` (1/2)

/java/MainActivity.java

```
public class MainActivity extends AppCompatActivity {
    @Override
    public void onCreate(Bundle savedInstanceState) {
        ...
        DrawerLayout drawer = findViewById(R.id.drawer);
        NavigationView nav = findViewById(R.id.nav_view);

        ActionBarDrawerToggle toggle =
            new ActionBarDrawerToggle(this, drawer, "Open", "Close");
        drawer.setDrawerListener(toggle);       toggle.syncState();
        getSupportActionBar().setDisplayHomeAsUpEnabled(true);
        ...
    }
    @Override
    public boolean onOptionsItemSelected(MenuItem item){
        if(toggle.onOptionsItemSelected(item)) return true;
    }
}
```


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DrawerLayout / NavigationView

6. Instancier le `DrawerLayout` (2/2)

/java/MainActivity.java

```
public class MainActivity extends AppCompatActivity
        implements NavigationView.OnNavigationItemSelectedListener {
    @Override
    public void onCreate(Bundle savedInstanceState) {
        ...
        FragmentManager fm = getSupportFragmentManager();
        nav.setNavigationItemSelectedListener(this);
        nav.getMenu().performIdentifierAction(R.id.nav_first, 0);
    }
    @Override
    public boolean onNavigationItemSelected(MenuItem item){
      if (item.getItemId() == R.id.nav_first){
          fm.beginTransaction().replace(R.id.contentFL,
                                        new FirstFragment()).commit();
      } else if (item.getItemId() == R.id.nav_second) { ... }
      drawer.closeDrawer(GravityCompat.START);
      return true;
    }
}
```


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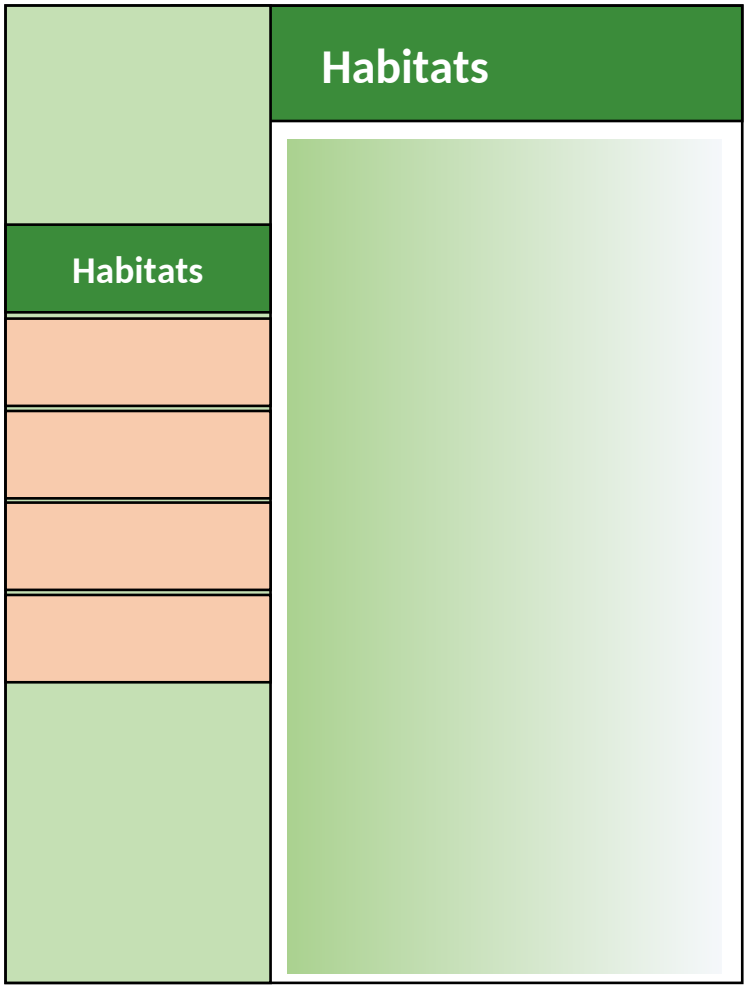
TP3b : Création d'un menu déroulant principal

Création d'un `DrawerLayout` avec un menu :

- **Habitats (`HabitatsFragment`)**
- **Mon habitat,**
- **Mes requêtes,**
- **Paramètres,**
- **Se déconnecter, …**



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